

Aritha Munasinghe

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EDUCATION

California State University, Sacramento

Sacramento, CA

Bachelor of Science in Computer Science

Expected Graduation: May 2027

- **GPA:** 4.0/4.0 (Dean's Honor List)
- **Relevant Coursework:** Data Structures & Algorithms, Computer Software Engineering, Applied Linear Algebra, System Programming (Unix), Computer Architecture.

EXPERIENCE

DRM Electronics

Colombo, Sri Lanka

Junior Software Engineer

Dec. 2023 – Aug. 2024

- Optimized **PostgreSQL** schemas and queries, reducing data retrieval latency by **40%** for reporting modules.
- Scaled **Node.js** backend architecture by refactoring APIs to handle high concurrent traffic with reliability.
- Modernized front-end in **Angular**, optimizing rendering to boost application responsiveness by **25%**.
- Built native Android prototypes (**Java**, **XML**) to validate cross-platform feasibility and user experience.

Software Engineering Intern

Aug. 2023 – Dec. 2023

- Developed responsive web interfaces using **HTML**, **CSS**, and **JavaScript**, optimizing layout consistency and usability to drive a **30%** increase in user engagement metrics.
- Engineered reusable **Angular** components to standardize UI patterns and accelerate feature deployment.
- Implemented **Node.js** backend services to optimize real-time data flow and support scalable API integrations.
- Collaborated in an Agile team using **Git** and **CI/CD** to ensure reliable, production-ready deployments.

PROJECTS

Generative AI Knowledge Retrieval Engine | *Python, FastAPI, PostgreSQL (pgvector), RAG*

Dec. 2025

- Engineered an end-to-end **RAG (Retrieval-Augmented Generation)** pipeline using **FastAPI** to enable context-aware semantic search over unstructured datasets.
- Implemented **PostgreSQL** with **pgvector** for high-dimensional embedding storage, enabling sub-second retrieval speeds for LLM grounding.
- Designed a modular data ingestion system that processes and chunks static files, transforming them into an interactive, queryable knowledge base.
- Optimized vector similarity search performance to minimize latency, ensuring a seamless and responsive “Chat with Data” user experience.

AI-Powered Campus Navigator | *Python, FastAPI, Tavily API, Data Pipelines*

Oct. 2025

- Engineered a centralized AI search engine using **FastAPI** and **OpenAI** that scrapes live web data to locate events, buildings, and resources in real-time.
- Built robust data pipelines using **Tavily API** to scrape and cache live events with sub-second latency.
- Developed a high-performance geospatial frontend with **React** and **TypeScript**, featuring an interactive map interface for visualizing dynamic event clusters.

Real-Time Collaborative Infrastructure | *React, TypeScript, WebSocket, Docker*

Aug. 2025

- Architected a low-latency collaboration platform using **React** and **Socket.IO**, optimizing event broadcasting to achieve sub-100ms synchronization for concurrent users.
- Designed a secure, sandboxed code execution environment using **Docker**, preventing malicious server-side operations while supporting runtime for 20+ languages.
- Orchestrated a robust CI/CD pipeline for automated delivery to Vercel and Railway, integrating **Supabase** for role-based session management.

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL), JavaScript (ES6+), TypeScript, Java, HTML/CSS, C

AI & Data: RAG Architectures, Vector Databases (pgvector), OpenAI API, Embeddings, Data Pipelines, Cassandra

Frameworks: FastAPI, React, Node.js, Angular, Socket.IO, Streamlit, NgRx

Cloud & Infrastructure: AWS (EC2, S3, Lambda), Docker, Git, CI/CD, Linux, Vercel, Railway

Analytics & Tools: Tableau, Power BI, Postman, VS Code, Agile Development, SDLC

Certifications: AWS Certified Solutions Architect – Associate (In Progress); Harvard CS50x: Intro to Computer Science